

MATH 2102 Linear Algebra II

Chapter 6

Connecting Operators and Matrices

Department of Mathematics, HKU

Chapter 6: Connecting Operators and Matrices

- A The overall picture
- B Bilinear forms
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A1. Operators vs matrices

In MATH2101 we studied linear algebra mainly around **matrices**. We discussed the null space, eigenvalues, eigenvectors, diagonalisability, characteristic polynomial, determinant(!), etc. of (often, square) matrices.

In this course, however, we have mainly focussed on the study of **operators**. We have likewise studied the null space (kernel), eigenvalues, eigenvectors, diagonalisability, characteristic polynomial etc. of operators.

While we use the same set of names in two apparently different contexts, how are they related to each other?

A1. Operators vs matrices

We know of course that each operator T on V can be represented by a square matrix A subject to a choice of basis of V . In short, operators are **basis-free** while matrices can be seen as **coordinate representations** of operators with respect to some choice of basis. Lot of questions then follow. For instance:

- Are the diagonalisability of T and the diagonalisability of A equivalent?
- Are the characteristic polynomial of T the same as the characteristic polynomial of A (given that their definitions are so different)?
- For the above two questions, as well as other questions of a similar nature, does the answer depend on the choice of basis?

The goal of this chapter is to somehow connect operator theory with matrix theory so as to give a big picture of what we have learned and answer these questions.

A2. Similarity invariant properties

In Chapter 2, we have seen that if $T \in \mathcal{L}(V)$ and A, B are the matrix representations of T with respect to two different bases of V , then

$$B = P^{-1}AP$$

where P is the change of basis matrix from the second basis to the first. In MATH2101, we have learned that such square matrices A and B are said to be **similar**.

A property of a matrix is said to be **similarity invariant** if the same property can be transferred to similar matrices. Some examples include invertibility and characteristic polynomial (in the sense defined in MATH2101 for matrices). The key point is that if a property is similarity invariant, then in connecting concepts between operators and matrices, we need not worry about the choice of basis.

A2. Similarity invariant properties

One of the central questions in the study of operators on V is to see whether we can find a basis of V consisting of eigenvectors of an operator T . If such basis exists, T is said to be **diagonalisable** since $\mathcal{M}(T)$ will be a diagonal matrix.

On the other hand, in MATH2101 we defined a square matrix A to be diagonalisable if there exists an invertible matrix P such that $P^{-1}AP$ is diagonal. Note that being diagonalisable is similarity invariant, and hence we see that the concepts of diagonalisability of operators and matrices are equivalent.

A2. Similarity invariant properties

Here are some other examples of correspondence between operators and matrices which do not depend on the choice of basis. Note that in such cases, the properties we know about such types of operators apply to the corresponding types of matrices, and vice versa, provided that we can associate the involved concepts between operators and matrices.

Invertible operators $\longleftrightarrow$ Invertible matrices

Nilpotent operators $\longleftrightarrow$ Nilpotent matrices

Idempotent operators $\longleftrightarrow$ Idempotent matrices


$$T^2 = T$$

A3. Eigenvalues and eigenvectors

Suppose $T \in \mathcal{L}(V)$ (where $\dim V = n$) and $A = \mathcal{M}(T)$ with respect to a fixed basis of V . We know that a scalar λ is said to be

- an eigenvalue of T if $T(\mathbf{v}) = \lambda\mathbf{v}$ for some non-zero $\mathbf{v} \in V$;
- an eigenvalue of A if $A\mathbf{x} = \lambda\mathbf{x}$ for some non-zero $\mathbf{x} \in \mathbb{F}^n$;

Are they equivalent? Is the concept of eigenvalues similarity invariant? How about eigenvectors?

A4. Jordan form of a matrix

A square matrix J is said to be in **Jordan (canonical/normal) form** if it has a block decomposition of the form

$$J = \begin{bmatrix} J_1 & O & \cdots & O \\ O & J_2 & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & J_k \end{bmatrix},$$

where each J_i is a square matrix such that

- all entries on the main diagonal are the same;
- all entries that are immediately above the main diagonal are 1; and
- all other entries are 0.

A4. Jordan form of a matrix

In Chapter 5, we have seen that when $\mathbb{F} = \mathbb{C}$, every operator on a vector space V has a Jordan basis. Translated into matrices, this says

For any matrix $A \in \mathbb{C}^{n \times n}$, there exists an invertible matrix P such that

$$P^{-1}AP = J$$

where J is a matrix in Jordan normal form. (We also say that J is the [Jordan form of \$A\$](#) .)

Note that each column of P is a generalised eigenvector of A .

A5. Correspondence via orthonormal basis

Not all concepts are similarity invariant. In Chapter 5, we have seen some connections between operators and matrices which depend on an **orthonormal basis**. The key bridge lies in the equality

$$\mathcal{M}(T^*) = (\mathcal{M}(T))^*.$$

which holds only if we have fixed orthonormal bases for the domain and codomain of T .

A5. Correspondence via orthonormal basis

Fix an orthonormal basis on an inner product space V . Suppose $T \in \mathcal{L}(V)$ and $A = \mathcal{M}(T)$. Then we know that

$$T \text{ is self-adjoint } (T^* = T) \iff A \text{ is Hermitian } (A^* = A)$$

$$T \text{ is normal } (T^* T = T T^*) \iff A \text{ is normal } (A^* A = A A^*)$$

$$T \text{ is unitary } (T^* = T^{-1}) \iff A \text{ is a unitary matrix } (A^* = A^{-1})$$

$$T \text{ is positive } (\langle T(\mathbf{v}), \mathbf{v} \rangle \geq 0) \iff A \text{ is positive semidefinite } (\langle A\mathbf{x}, \mathbf{x} \rangle \geq 0)$$

A5. Correspondence via orthonormal basis

Results about orthogonal diagonalisability of operators then translate to the corresponding diagonalisability for matrices. For instance, we know from Chapter 5 that

For every normal operator $T \in \mathcal{L}(V)$, there exists an orthonormal basis of V consisting of eigenvectors of T .

We thus have a corresponding result in the context of matrices:

For every normal matrix A , there exists a unitary matrix Q and a diagonal matrix D such that $A = QDQ^*$.

A6. The missing piece

So far in this chapter we have seen many examples of how concepts about operators that we learnt in this course correspond very well to similar concepts about matrices, many of which we have already learnt in MATH2101.

However, a missing piece remains. We have not been able to connect the characteristic polynomial of an operator, which was defined in Chapter 5, with the characteristic polynomial of a matrix, which we learnt in MATH2101. The latter involves the concept of **determinants**, which are (so far) defined on (square) matrices only. We shall devote the rest of this chapter to connect this final piece.

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B1. Setting the scene

For the rest of this chapter, we assume that V is a vector space over $\mathbb{F}$ with dimension $n > 0$, unless otherwise specified.

B2. Motivation

Eventually we want to define the concept of determinants in a way that bridges operator theory and matrix theory. In a sense, the determinant (of an $n \times n$ matrix) can be considered as a function

$$\det : \underbrace{\mathbb{F}^n \times \mathbb{F}^n \times \cdots \times \mathbb{F}^n}_{n \text{ times}} \rightarrow \mathbb{F}.$$

Recall that a linear function $f : V \rightarrow \mathbb{F}$ is called a **linear functional**. Hence the determinant somehow looks like a ‘high-dimensional version’ of linear functionals.

B3. Definition

Recall from Chapter 4 that on an inner product space, the maps

$$\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u} \rangle \quad \text{and} \quad \mathbf{v} \mapsto \langle \mathbf{u}, \mathbf{v} \rangle$$

(where $\mathbf{u}$ is fixed) are quite different, as the former is linear but the latter is not.

We define a **bilinear form** on V to be a function $\beta: V \times V \rightarrow \mathbb{F}$ such that

$$v \mapsto \beta(\mathbf{v}, \mathbf{u}) \quad \text{and} \quad v \mapsto \beta(\mathbf{u}, \mathbf{v})$$

are both linear functionals on V for any fixed choice of $\mathbf{u}$. (In this sense, **bilinear functional** would have been a better name although it is much less popular.)

B3. Definition

Examples of bilinear forms:

- On $\mathbb{F}^3$, define

$$\beta(\mathbf{x}, \mathbf{y}) = x_1y_2 + 2x_2y_3 + 3x_3y_1 + 4x_3y_3.$$

- More generally, let $A \in \mathbb{F}^{n,n}$ with (i,j) -entry a_{ij} . On $\mathbb{F}^n$, define

$$\beta(\mathbf{x}, \mathbf{y}) = \sum_{i,j} a_{ij}x_iy_j.$$

- On $P_3(\mathbb{R})$, define $\beta(p, q) = p(2)q'(3)$.
- On arbitrary V , take linear functionals $f_1, f_2, f_3, \dots$ and $g_1, g_2, g_3, \dots$ on V , and define

$$\beta(\mathbf{u}, \mathbf{v}) = f_1(\mathbf{u})g_1(\mathbf{v}) + f_2(\mathbf{u})g_2(\mathbf{v}) + \cdots + f_k(\mathbf{u})g_k(\mathbf{v}).$$

B4. The vector space $V^{(2)}$

The set of all bilinear forms on V form a vector space over $\mathbb{F}$, with the zero map as the zero vector and usual addition and scalar multiplication of functions. We denote this space by $V^{(2)}$.

Fix a basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ of V . Then β is completely determined by the values of $\beta(\mathbf{e}_i, \mathbf{e}_j)$ for all i, j . We use $\mathcal{M}(\beta)$ to denote the $n \times n$ matrix whose (i, j) -entry is $\beta(\mathbf{e}_i, \mathbf{e}_j)$, and call it the **matrix representation of β (with respect to that basis)**.

With this, we see that

The dimension of $V^{(2)}$ is $(\dim V)^2 = n^2$.

B5. Bilinear forms and linear maps

Let $\beta \in V^{(2)}$ and $T \in \mathcal{L}(V)$. It is easy to check that

$$\alpha(\mathbf{u}, \mathbf{v}) = \beta(\mathbf{u}, T(\mathbf{v})) \quad \text{and} \quad \gamma(\mathbf{u}, \mathbf{v}) = \beta(T(\mathbf{u}), \mathbf{v})$$

are both in $V^{(2)}$. Furthermore, we have

$$\mathcal{M}(\alpha) = \underline{\mathcal{M}(\beta)\mathcal{M}(T)} \quad \text{and} \quad \mathcal{M}(\gamma) = \underline{\mathcal{M}(T)^T \mathcal{M}(\beta)}$$

B5. Bilinear forms and linear maps

Change-of-basis formula

Suppose $\beta \in V^{(2)}$ and let A, B be the matrix representation of β with respect to bases $\mathcal{A}$ and $\mathcal{B}$ respectively. Let C be the change of basis matrix from $\mathcal{A}$ to $\mathcal{B}$. Then

$$A = \underline{C^T B C}.$$

B5. Bilinear forms and linear maps

As an illustration, consider the bilinear form $\beta(p, q) = p(2)q'(3)$ on $P_3(\mathbb{R})$, with $\mathcal{A} = \{1, x - 2, (x - 3)^2\}$ and $\mathcal{B} = \{1, x, x^2\}$.

$$M(\beta, \mathcal{A}) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad M(\beta, \mathcal{B}) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 2 & 4 \\ 6 & 12 & 24 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$2x - 6$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 6 \end{bmatrix}$$

B6. Special types of bilinear forms

A bilinear form $\beta \in V^{(2)}$ is said to be

- **diagonalisable** if, with respect to some basis of V , $\mathcal{M}(\beta)$ is a diagonal matrix;
- **symmetric** if $\beta(\mathbf{u}, \mathbf{v}) = \beta(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in V$;
- **alternating** if $\beta(\mathbf{v}, \mathbf{v}) = 0$ for all $\mathbf{v} \in V$.

Give examples.

$$\alpha(\mathbf{u}, \mathbf{w}) = -\alpha(\mathbf{w}, \mathbf{u})$$

proof: $0 = \alpha(\mathbf{u} + \mathbf{w}, \mathbf{u} + \mathbf{w})$

$$= \alpha(\mathbf{u}, \mathbf{u}) + \alpha(\mathbf{u}, \mathbf{w}) + \alpha(\mathbf{w}, \mathbf{u}) + \alpha(\mathbf{w}, \mathbf{w})$$

$$= \alpha(\mathbf{u}, \mathbf{w}) + \alpha(\mathbf{w}, \mathbf{u}) = 0$$

B6. Special types of bilinear forms

We denote

- the set of all symmetric bilinear forms on V by $V_{\text{sym}}^{(2)}$;
- the set of all alternating bilinear forms on V by $V_{\text{alt}}^{(2)}$.

It is easy to check that $V_{\text{sym}}^{(2)}$ and $V_{\text{alt}}^{(2)}$ are subspaces of $V^{(2)}$.

B6. Special types of bilinear forms

A bilinear form $\beta \in V^{(2)}$ is diagonalisable if and only if it is symmetric.

Proof:

$\nearrow \{e_1, \dots, e_n\}$

($\Rightarrow$) Fix a basis of V for which $\mathcal{M}(\beta)$ is diagonal. Then $\beta(\mathbf{u}, \mathbf{v}) = \beta(\mathbf{v}, \mathbf{u})$ since

$$\beta \left(\sum_{i=1}^n v_i e_i, \sum_{j=1}^n u_j e_j \right)$$

B6. Special types of bilinear forms

A bilinear form $\beta \in V^{(2)}$ is diagonalisable if and only if it is symmetric.

Proof:

($\Leftarrow$) We prove by induction on n . The result holds for $n = 1$. Suppose $n > 1$ and the result holds for smaller n . Pick $\mathbf{v} \in V$ such that $\beta(\mathbf{v}, \mathbf{v}) \neq 0$. Consider

$$U = \{\mathbf{u} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0\}.$$

Then U is subspace of V with dimension _____.

The following propositions are equivalent:

① p is symmetric

② $M(p)$ is symmetric to all basis of V

③ $M(p)$ is symmetric to some basis of V

④ $M(p)$ is diagonalizable to some basis of V .

① $\rightarrow$ ③. $\exists \{e_i\} \quad p(e_i, e_j) = p(e_j, e_i) \rightarrow M(p)_{i,j} = M(p)_{j,i}$

③ $\rightarrow$ ②. obviously

④ $\rightarrow$ ①: Suppose $M(p, \{e_i\})$ is symmetric.

$$\begin{aligned} p(u, w) &= p(\sum u_i e_i, \sum w_j e_j) \\ &= \sum \sum u_i w_j p(e_i, e_j) \\ &= p(\sum w_j e_j, \sum u_i e_i) \\ &= p(w, u) \rightarrow \text{①.} \end{aligned}$$

① $\rightarrow$ ④: We induction.

base case: when $n=1$, all matrix are diagonal matrix.

suppose p symmetric $\Rightarrow p$ diagonalizable holds for the cases where dimension is one less than the case we want to prove.

suppose p is symmetric. if $p=0$, then $\sim$, so suppose $p \neq 0$.

$$\therefore \exists u, w, \text{ s.t. } 0 \neq p(u, w) = \frac{1}{2} (p(u+w, u+w) - p(u, u) - p(w, w))$$

$$\therefore \exists v. p(v, v) \neq 0.$$

$$\text{let } U = \{u \in V : p(u, v) = 0\} \rightarrow \dim U = n-1.$$

$$M(p, \{u_1, u_2, \dots, u_{n-1}, v\}) \rightarrow \sim.$$

B6. Special types of bilinear forms

$$V^{(2)} = V_{\text{sym}}^{(2)} \oplus V_{\text{alt}}^{(2)}.$$

Proof:

We have

$$\beta(\mathbf{u}, \mathbf{v}) = \frac{\beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{v}, \mathbf{u})}{2} + \frac{\beta(\mathbf{u}, \mathbf{v}) - \beta(\mathbf{v}, \mathbf{u})}{2}.$$

$= \frac{1}{2}(\beta(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{v}) - \beta(\mathbf{u}, \mathbf{u}) - \beta(\mathbf{v}, \mathbf{v})) = 0$

On the other hand, if β is both ~~sym~~metric and alternating, then

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C1. Definition

An **m -linear form** on V is a function $\beta: V^m \rightarrow \mathbb{F}$ that is linear in each slot when the others are held fixed. A **multilinear form** is an m -linear form for some m .

Some remarks:

- A 1-linear form is a L^T .
- A 2-linear form is a bilinear.
- The set $V^{(m)}$ of all m -linear forms on V is a vector space.

C2. Alternating multilinear forms

An m -linear form β is said to be **alternating** if

$$\beta(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m) = 0 \quad \text{whenever at least two of the } \mathbf{v}_i \text{ are equal.}$$

The set of all alternating m -linear forms on V is denoted by $V_{\text{alt}}^{(m)}$. It can be checked that $V_{\text{alt}}^{(m)}$ is a subspace of $V^{(m)}$.

C2. Alternating multilinear forms

Let $\beta \in V_{\text{alt}}^{(m)}$. Then

$\beta(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m) = 0$ whenever $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ are linearly dependent.

Hence $V_{\text{alt}}^{(m)} = \{0\}$ when $m > n$.

$$\text{let } v_k = \sum_{i=1}^k \lambda_i v_i$$

$$\beta(v_1, \dots) = \beta(v_1, \dots, v_k, \sum \lambda_i v_i, \dots)$$

$$= \sum \beta(\dots, v_k, v_j, \dots) = 0$$

C2. Alternating multilinear forms

Let $\beta \in V_{\text{alt}}^{(m)}$. Then $\beta(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$ changes sign when two of the $\mathbf{v}_i$'s are swapped.

$$\beta(\dots, v_i, v_j, \dots, v_j, v_i, \dots) = 0 = \dots$$

like bilinear.

C3. Alternating n -linear forms

Suppose $\alpha \in V_{\text{alt}}^{(n)}$. If we fix a basis of V , then every $\mathbf{v} \in V$ can be expressed as a coordinate vector with respect to that basis. Hence each time the input to α can be viewed as an $n \times n$ matrix.

C3. Alternating n -linear forms

If $\alpha \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1$, can you find the following?

(a) $\alpha \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix} = 6$

(b) $\alpha \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

Handwritten notes in red ink:

$$\begin{array}{ccc|c} 1 & 1 & 1 & \\ 0 & 2 & 2 & + \\ 0 & 0 & 0 & \end{array}$$

$$\begin{array}{ccc} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array}$$

C3. Alternating n -linear forms

We shall not go into the technical proof of the following result, but the idea should be clear from the example we have just seen.

The space $V_{\text{alt}}^{(n)}$ is 1-dimensional. In particular, suppose $\alpha \in V_{\text{alt}}^{(n)}$. If we fix a basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ of V and set $\alpha(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = 1$, then we have

$$\alpha(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = \sum_{\sigma \in S_n} (\text{sgn } \sigma) b_{\sigma(1),1} b_{\sigma(2),2} \cdots b_{\sigma(n),n}.$$

where b_{ij} denotes the coefficient of $\mathbf{e}_i$ when $\mathbf{v}_j$ is expressed as a linear combination of the basis vectors.

C3. Alternating n -linear forms

Let $\alpha \in V_{\text{alt}}^{(n)}$ be nonzero. Then

$\alpha(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) \neq 0$ whenever $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent.

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D1. Definition

Let $T \in \mathcal{L}(V)$ and $\alpha \in V_{\text{alt}}^{(n)}$. Define $\alpha_T : V^n \rightarrow \mathbb{F}$ by

$$\alpha_T(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = \alpha(T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)).$$

It is not hard to check that

- $\alpha_T \in V_{\text{alt}}^{(n)}$;
- the map $\alpha \mapsto \alpha_T$ is linear.

Since $V_{\text{alt}}^{(n)}$ has dimension 1, α_T must be a scalar multiple of α . Hence we have the following definition.

The **determinant** of an operator T , denoted $\det T$, is the unique scalar such that

$$\alpha_T = (\det T)\alpha \quad \text{for every } \alpha \in V_{\text{alt}}^{(n)}.$$

D1. Definition

We can also define the determinant of a matrix as follows.

Let $A \in \mathbb{F}^{n \times n}$. We define the determinant of A by

$$\det A = \det T,$$

where T is the operator on $\mathbb{F}^n$ defined by $T(\mathbf{x}) = A\mathbf{x}$.

D2. Examples

Suppose $\lambda \in \mathbb{F}$, $T \in \mathcal{L}(V)$. From the definition, it is easy to see the following.

- The identity map, as well as the identity matrix, has determinant 1.
- The map λI on V has determinant λ^n .
- $\det \lambda T = \lambda^n \det T$.
- If there is a basis of V containing eigenvectors of T , which correspond to eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, then $\det T = \prod \lambda_i$.
- The determinant of a diagonal matrix is equal to the product of its diagonal entries.

D3. Formula

It is easy to see that the determinant can be seen as an alternating n -linear form on $\mathbb{F}^n$. Hence we have the following.

Let A be an $n \times n$ matrix whose (i, j) -entry is a_{ij} . Then

$$\det A = \sum_{\sigma \in S_n} (\operatorname{sgn} \sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}.$$

D3. Formula

The formula readily explains certain properties, such as the following.

- The ‘diagonal multiplication’ method can be used to evaluate 2×2 and 3×3 determinants, but fails for 4×4 or above.
- The determinant of a triangular matrix is equal to the product of its diagonal entries.

D4. Properties

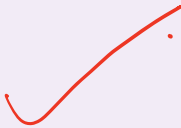
For any $S, T \in \mathcal{L}(V)$, we have

$$\begin{aligned} \alpha_{ST} &= \alpha(STv \cdot) \\ &= \det S \alpha(Tv \cdot) \\ &= \det S \det T \alpha(v \cdot) \end{aligned}$$

$$\det ST = \det S \det T. \quad \dots$$

It follows that $\det AB = \det A \det B$ for square matrices A, B of the same size.

D4. Properties

- An operator $T \in \mathcal{L}(V)$ is invertible if and only if $\det T \neq 0$.
 - When this is the case, we have $\det T^{-1} = \frac{1}{\det T}$.
 - The same holds for matrices in place of operators.
 - Determinant is thus similarity invariant.
- 

D4. Properties

For any square matrix A , we have $\det A = \det A^T$.

D4. Properties

Most other properties about determinants that you learnt in MATH2101 can be verified using our (new?) definition. Some examples include:

- Cofactor expansion theorem
- A square matrix with two identical columns has zero determinant
- Effect of EROs on determinant

+ 9C - $\frac{1}{2}$

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E1. Trace



We define the **trace** of a square matrix A , denoted $\text{tr } A$, to be the sum of its diagonal entries. It is easy to check that

$$\text{tr } AB = \text{tr } BA$$

for any square matrices A and B of the same size. It follows that

- trace is similarity invariant;
- we can define the trace of an operator to be the trace of any of its matrix representation.

$$\mathbb{F} = \mathbb{C}, \quad \text{tr } A = \sum m_i \lambda_i$$

↓
multiplicity

$$8.02 - 8.57$$

E2. More on determinants of operators

For any $S \in \mathcal{L}(W, V)$ and $T \in \mathcal{L}(V)$, we have

$$\det(S^{-1}TS) = \det T$$

whenever S^{-1} exists.

E2. More on determinants of operators

$$\det T = \det M(E^{-1}TE)$$

For any $T \in \mathcal{L}(V)$, we have $\det T = \det M(T)$ regardless of the choice of basis. It follows that the dual map T' has the same determinant as T .

E3. Sum and product of eigenvalues

Suppose $T \in \mathcal{L}(V)$ and $\mathbb{F} = \mathbb{C}$. Then, counting multiplicities,

- the sum of the eigenvalues of T is equal to $\text{tr } T$;
- the product of the eigenvalues of T is equal to $\det T$.

E4. Determinant on inner product space

Let T be an operator on an inner product space V .

(1) $\det T^* = \overline{\det T}$. $\rightarrow \det T \overline{\det T} = 1$

(2) If T is unitary, $\det T = \pm 1$.

(3) If T is positive, $\det T \geq 0$.

E5. Singular values

Let T be an operator on an inner product space V . Then the product of the singular values of T is equal to $|\det T|$.

Proof:

It suffices to show that $\det(T^* T) = s_1^2 s_2^2 \cdots s_n^2$, where $s_1, s_2, \dots, s_n$ are the singular values of T .

E6. Characteristic polynomial redefined

Given an operator $T \in \mathcal{L}(V)$, we (re)define the **characteristic polynomial** of T , denoted $\chi_T(t)$, by

$$\chi_T(t) = \det(tI - T).$$

In what sense is this a generalisation of the previous definition?

E6. Characteristic polynomial redefined

Cayley-Hamilton Theorem

For any $T \in \mathcal{L}(V)$, we have $\chi_T(T) = 0$.

Proof:

We may assume $\mathbb{F} = \mathbb{R}$. Let A be the matrix of T with respect to a certain basis. Define $S \in \mathcal{L}(\mathbb{C}^n)$ by $S(\mathbf{x}) = A\mathbf{x}$.

E6. Characteristic polynomial redefined

For any $T \in \mathcal{L}(V)$, the characteristic polynomial of T is of the form

$$\chi_T(t) = t^n - (\operatorname{tr} T)t^{n-1} + \cdots + (-1)^n(\det T).$$